

Timeline	Stage	Key Activities & Deliverables
Phase 1 — Hardware, Simulation & Manufacturing Implementation		Months 9–1
September (9)	Research & V-Model Formulation	- Execute thorough literature review and resource collection. - Apply the V-Model framework to identify solid requirements, precise mechatronic specifications, and the core features to focus on.
October (10)	Conceptual Design	- Draft the conceptual mechanical design for the platform's physical structure, ensuring modularity of the base, torso, arms, and head. - Engineer the initial battery pack and map out the power distribution design. - Outline custom circuit designs and architect the low-level embedded control systems.
November (11)	Final Design	- Apply Design for Manufacturing (DFM) and Design for Assembly (DFA) principles to optimize physical fabrication and integration. - Lock in the mechanical structure with a strict focus on modularity so the base, torso, arms, and head support independent development and future upgrades. - Finalize the exact PCB designs and precise trace routing to support the distributed electrical and embedded-control architecture. - Complete the final battery pack design to guarantee stable power delivery under dynamic, high-torque loads. - Engineer robotic thermal management and cooling systems to protect high-load AI processors and joint actuators from overheating. - Execute a comprehensive component availability check to ensure all specified electronics and hardware can be procured without supply chain delays. Procurement phase officially begins at the close of Final Design, feeding directly into the manufacturing track below.
Dec & Jan (12 & 1)	High-Level Software Development & Full Simulation	- Develop the high-level software stack: autonomous navigation and arm control. - Construct a full, physics-accurate simulation environment and validate the high-level software within it. - Manufacturing runs in parallel — procurement, fabrication, and assembly (see Manufacturing Track below).
Manufacturing Track (12 → 2)	Parallel Fabrication & Assembly	- Runs in parallel with all Dec–Feb software and dashboard workstreams. - Procure components, fabricate parts, and assemble the robotic platform.
Phase Deliverable — Finalized CAD and PCB designs, a complete integration design, manufacturing progress validated against the simulation environment, and first-stage high-level software (autonomous & arm control) working in simulation.		

Phase 2 — Software Integration, Hardware Validation & AI Development

Months 2–6

February (2)	Dashboard Development & High-Level Software Finalization	• Begin development of the user dashboard and interface. • Finalize the high-level software (autonomous navigation and arm control) based on simulation results. • Manufacturing (fabrication & assembly) concludes in parallel with these tasks.
March (3)	Dashboard Continuation & Low-Level Hardware Validation	• Continue development of the user dashboard and interface. • Validate the physical hardware against the low-level embedded software.
April (4)	Hardware Validation & Task Port Integration	• Validate the physical hardware against the high-level software (autonomous navigation & arm control). • Establish a single, standardized command interface for high-level orders and integrate the task port. • Validate the dedicated software port by successfully flashing a task-specific proof of concept.
May & June (5 & 6)	Task Port Verification, Final Verification & Additional Features	• Execute comprehensive task port verification and overall proof-of-working / concept testing to validate the platform end-to-end. • Explore additional features where scope allows — Vision-Language-Action (VLA) model and further AI integration.

Phase Deliverable — Fully integrated robotic platform, hardware validated against both high-level and low-level software, verified end-to-end against all functional requirements; additional AI-driven capabilities (VLA / AI integration) explored as stretch goals, scope permitting.